

Organizational Knowledge Gap Intelligence Platform

Module-wise presentation prep — 5 speakers · 10 minutes · 12 modules

One-line pitch: A full-stack platform that finds expertise gaps across an organization, benchmarks employees against role-specific competency frameworks, and closes gaps through AI-generated learning paths, mentorship, and analytics dashboards.

Stack: Java 17 · Spring Boot 3.3 · Spring Security (JWT + OAuth2) · Spring Data JPA · PostgreSQL | React 18 · Vite · Recharts | OpenAI/Gemini · OpenPDF · Apache POI · Docker

Speaker 1 - Foundation & Identity

~2 min

Module 1 · Authentication & RBAC

PROBLEM

Organizations need secure, role-appropriate access - a single login can't expose HR data to every employee.

KEY FEATURES

JWT login, OAuth2 (Google + GitHub), password-reset OTP over email, 6 roles with fine-grained access control.

■ DEMO TIP

Log in with Google, then show the same page rendering differently for an Employee vs an Admin.

Module 2 · Employee Profile & Skill Inventory

PROBLEM

There is no single source of truth for who knows what - skills live in people's heads, not in the system.

KEY FEATURES

Rich profiles, skill self- & peer-rating, certifications, and work-experience tracking.

■ DEMO TIP

Open a seeded profile and add a skill rating live to show the inventory updating.

Speaker 2 - Benchmarking & Gap Analysis

~2 min

Module 3 · Competency Framework & Role Benchmarking

PROBLEM

Skills mean nothing without a bar to measure against - every role needs a defined 'good'.

KEY FEATURES

Admins define required skill levels per role and department as reusable competency frameworks.

■ DEMO TIP

Show a framework for one role and point out the required vs actual levels.

Module 4 · Knowledge Gap Analysis

PROBLEM

Managers can't see where teams fall short of role requirements, so training becomes guesswork.

KEY FEATURES

Individual / team / department gap scoring with severity, a skill-gap heatmap, and trend-over-time.

■ DEMO TIP

This is the flagship - open the Org Gap Dashboard heatmap; it's the strongest visual in the demo.

Speaker 3 - Growth: Learning, Mentorship, Progress

~2 min

Module 5 · Training Recommendation

PROBLEM

Knowing a gap exists doesn't fix it - people need a concrete, personalized path forward.

KEY FEATURES

AI-generated learning paths (OpenAI / Gemini) with curated external resource links, mapped to each gap.

■ DEMO TIP

Click 'recommend' on a real gap and let the AI generate a learning path on screen.

Module 6 · Knowledge-Sharing & Mentorship

PROBLEM

Expertise stays siloed; employees don't know who to learn from inside their own company.

KEY FEATURES

Expert directory, mentorship session booking, and a reusable knowledge-article library.

■ DEMO TIP

Book a mentorship session with a seeded expert to show the flow end-to-end.

Module 7 · Learning Progress Tracking

PROBLEM

Enrolling in training is easy; finishing it and proving impact is where organizations lose track.

KEY FEATURES

Enrollments, milestones, certification-expiry reminders, and learning-velocity analytics.

■ DEMO TIP

Show the Learning Analytics page - velocity chart plus a milestone being completed.

Speaker 4 - Measurement & Insight

~2 min

Module 8 · Assessment & Survey

PROBLEM

Self-reported skills are biased - you need objective and multi-source evidence of ability.

KEY FEATURES

Skill quizzes, 360 self/peer/manager reviews, and historical score comparison.

■ DEMO TIP

Take a short quiz, submit, then open Assessment History to show the score trend.

Module 9 · Notifications

PROBLEM

Insights are useless if no one acts on them - gaps and deadlines get forgotten.

KEY FEATURES

Gap alerts, deadline & session reminders, milestone achievements - email + in-app + a daily scheduler.

■ DEMO TIP

Open the notification bell and show an unread gap alert; mention the daily scheduler job.

Module 10 · Analytics Dashboards

PROBLEM

Different roles need different views - an employee and an HR head care about different numbers.

KEY FEATURES

Tailored Employee, Manager, and HR/Admin dashboards with org-wide KPIs.

■ DEMO TIP

Switch between the Employee dashboard and the Admin dashboard to show role-tailored KPIs.

Speaker 5 - Delivery: Reports, Deployment, Close

~2 min

Module 11 · Reports & Export

PROBLEM

Leadership decisions and audits need shareable artifacts, not a screen someone has to log into.

KEY FEATURES

Individual gap report as PDF; department / workforce / training-effectiveness / ROI / strategic reports as Excel.

■ DEMO TIP

Export an individual gap PDF live and open the downloaded file to prove it works.

Module 12 · Integration, Testing & Deployment

PROBLEM

A demo that only runs on one laptop isn't a product - it must be reproducible and tested.

KEY FEATURES

Dockerized full stack, JUnit 5 + Mockito + Vitest tests, Postman collection, one-command run.

■ DEMO TIP

Show 'docker compose up' bringing up DB + backend + frontend, then wrap with the roadmap.

Delivery tips for a tight 10 minutes

- Budget ~2 min per speaker. Lead with the **problem**, then one feature, then the demo - don't read the slide.
- Modules 4 (Gap heatmap) and 5 (AI recommendation) are your wow moments - give them the most time.
- Have the app pre-seeded and logged in before you start; never wait for a build during the talk.
- One person drives the screen while each speaker talks, so transitions stay seamless.
- Close on Module 12: it proves the product is real, reproducible, and tested - then invite questions.